

Guardrail Cascades Do Not Compose: Measuring Routing-Induced Starvation in a Three-Layer Prompt-Injection Defense

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Abstract—Prompt injection and jailbreak attacks remain the dominant failure mode for deployed large language model (LLM) systems, and practitioners increasingly stack several detectors—a fast surface classifier, a hidden-state probe, and an output auditor—to defend in depth. The implicit assumption behind such stacks is that layer-level accuracy *composes*: a stronger component yields a stronger system. We stress-test this assumption on a three-layer, fail-closed guardrail wrapped around Qwen2.5-7B-Instruct on two T4 GPUs. In isolation, our hidden-state probe attains 0.997 ROC-AUC on a held-out injection set; yet when routed through a conventional escalation gate that forwards only “suspicious” inputs to the second layer, the full pipeline’s recall (0.37) is statistically indistinguishable from that of the surface classifier alone (0.35). The gate silently starved the probe of 38 of 60 held-out injections. Removing the gate—running the second layer always-on—raises end-to-end recall to 0.90 at a 1.8% false-positive rate, at the cost of one additional 7B forward pass (~50–150 ms) per request. We formalize why this occurs, show that component-level metrics do not predict ensemble security, and distill routing-policy recommendations for layered LLM guardrails. Code, artifacts, and seeds are released for reproduction on free-tier hardware.

Index Terms—prompt injection; jailbreak; LLM security; defense in depth; hidden-state probes; guardrail evaluation; cascade classifiers

I. INTRODUCTION

Prompt injection—where an attacker smuggles instructions into the context window of an LLM to hijack its goal or exfiltrate data—is widely ranked as a leading security risk for LLM applications [3]. The attack class spans direct instruction overrides (“ignore your previous instructions”) [1], indirect injection through retrieved or tool-supplied content [2], [3], and jailbreaks that override safety alignment [4]. Because no single detector is robust to all of these, modern guardrail practice is defense in depth: a cheap, high-precision surface classifier screens inputs; a second, more expensive mechanism such as a hidden-state probe or an LLM-based judge re-examines the remainder; and output auditors check the response after generation [5]–[9].

Each layer of such a stack is usually developed, benchmarked, and reported *in isolation*, and the implicit expectation is that a layer with strong standalone metrics will strengthen the ensemble. In this paper we show that this expectation can fail in a precisely quantifiable way. We build and release a three-layer, fail-closed injection guard around Qwen2.5-7B-Instruct [16], reproducible on two T4 GPUs, and evaluate it end-to-end on a held-out injection corpus. The second layer—a logistic probe trained on the intermediate hidden states of the target

LLM—achieves 0.997 ROC-AUC and 0.783 recall *in isolation*. Yet when this probe is placed behind the standard “escalation gate” that only forwards inputs the surface classifier finds suspicious, the end-to-end recall of the full pipeline (0.37) collapses to that of the surface classifier alone (0.35): the gate forwarded only 1 of 60 held-out injections to the probe. The probe, despite its near-perfect AUC, contributed essentially one detection to the ensemble.

This is an instance of a classic principle in cascade design: a cheap first stage in a cascade must have near-total recall, or every stage behind it is starved of the very traffic it exists to catch [13]. Our contribution is not the discovery of this principle, but the demonstration that production-style LLM guardrail stacks violate it silently, together with a quantified cost of the violation and a concrete fix. Specifically, we contribute: (i) a reproducible, fail-closed three-layer prompt-injection pipeline for Qwen2.5-7B-Instruct on two T4 GPUs, released as open source; (ii) a quantified *gate-starvation* result—a component with 0.997 AUC contributes a single detection end-to-end because the routing gate inherits the surface classifier’s blind spots, cutting end-to-end recall from the component’s 0.78 to 0.37; (iii) a minimal, costed fix—always-on routing of the second layer—which restores end-to-end recall to 0.90 at 1.8% FPR for one extra 7B forward pass per request, together with a threshold operating-point analysis; and (iv) routing-policy recommendations and a discussion of the methodological traps (small test splits, in-sample threshold calibration, augmentation leakage) that hide this failure mode from component-level benchmarks.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the pipeline. Section IV details data, training, and evaluation protocol. Section V reports results, including the central routing pathology. Section VI discusses implications, limitations, and recommendations, and Section VII concludes.

II. BACKGROUND AND RELATED WORK

A. THREAT MODEL

We consider a single-turn LLM application that (i) accepts a user instruction, (ii) optionally augments it with *untrusted* context (retrieved documents, tool output, web content), and (iii) generates a response under a fixed, sensitive system prompt. The adversary controls either the user instruction (direct injection) or the untrusted context (indirect injection), and seeks goal hijacking or system-prompt exfiltration. This matches the canonical formulations of Perez and Ribeiro [1] and Greshake et al. [2], and the indirect-injection surface that dominates current agentic deployments [2], [3].

B. DETECTION DEFENSES

Two families of detector dominate practice. *Surface prompt guards* are lightweight classifiers—typically fine-tuned DeBERTa-family encoders—that score raw input text. Examples include prompt guard models trained as binary classifiers, such as PIGuard [6] and InjecGuard [7], and generative guards such as Llama Guard [5]. These models are fast and cheap but are documented to suffer *over-defense*: bias toward trigger words causes benign inputs stuffed with attack vocabulary to be misclassified, with accuracy dropping toward chance on adversarial-benign sets such as NotInject [6], [7]. We observe the same failure mode in our first layer, which motivates treating it as a high-precision but recall-limited stage.

Representation-level detectors read the model’s internal activations rather than the surface string. Linear probes on intermediate hidden states have a long lineage [12] and a growing body of recent work for injection and jailbreak detection: InstructDetector uses hidden states and gradients to detect indirect instructions [8]; JBShield analyzes activated toxic/jailbreak concepts in hidden representations [9]; and large-scale evaluations train activation probes across many corpora and argue for leave-one-dataset-out evaluation [10], [11]. These methods are generally more robust to surface obfuscation than string-level guards, at the cost of a full forward pass of the target LLM.

C. CASCADES AND DEFERRAL

Ordering a cheap high-recall stage before an expensive high-precision stage is the classic cascade pattern [13], generalized in the machine learning literature as *learning to defer* [14]. The key invariant—established decades ago and restated here because it is the mechanism of our central finding—is that no stage can detect an input that a prior stage refused to pass on. In Section VI-A we formalize this as an upper bound on ensemble recall and show that our guardrail stack sits exactly at that bound.

III. SYSTEM DESCRIPTION

The pipeline (Fig. 1) wraps a target LLM in three defensive layers, summarized in Table I. Every stage is *fail-closed*: a crash, an out-of-memory error, an empty judge verdict, or a mismatched probe artifact blocks the request rather than passing it. All layers share a single input format—untrusted context is wrapped in explicit `<untrusted_context>` delimiters—so training, scoring, and generation operate on the same distribution.

TABLE I THE THREE DEFENSIVE LAYERS OF THE PIPELINE

Layer	Component	Function	Placement
L1	ProtectAI DeBERTa-v3 (v2), optional PIGuard ensemble	Sliding-window surface audit; high precision, recall-limited	GPU 1
	Logistic probe on LLM hidden states (layer 20)	Behavioral recall engine; catches obfuscated/implicit instructions	
L2	Canary token + regex heuristics + self-judge	Post-generation output audit (exfiltration, compromise)	reuses the 7B

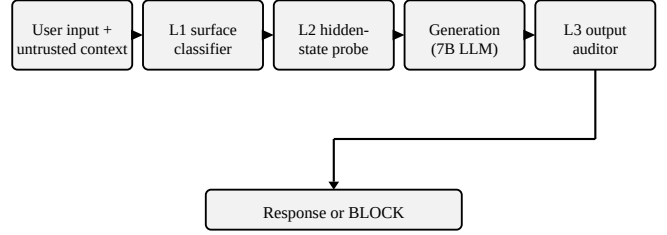


Fig. 1. The three-layer pipeline. L2 runs on every input L1 does not hard-block (“always-on” routing); L3 audits the generated output. Any layer can block the request, fail-closed.

A. LAYER 1 — SURFACE CLASSIFIER

L1 audits raw text with a fine-tuned ProtectAI/DeBERTa-v3-base-prompt-injection-v2 classifier [15]. Inputs longer than the encoder window are split into overlapping sliding windows with two production guards: an *overlap clamp* (overlap capped at a quarter of the window budget) and a *window cap* (at most 64 windows, always including the document tail), so a long RAG document cannot silently hide an attack or explode into hundreds of classifier calls. A single-window input is labeled INJECTION at probability at least 0.5; a multi-window input requires either 2 or more alert windows or a peak of at least 0.95, so one quoted attack inside a long benign document escalates rather than blocks. An optional ensemble (e.g., adding PIGuard [6]) votes per window, diluting single-model quirks such as repetition-triggered false spikes.

B. LAYER 2 — HIDDEN-STATE PROBE

L2 is a low-capacity logistic regression trained on the intermediate activations of the *target* LLM. For Qwen2.5-7B-Instruct (28 layers) we extract the last-token hidden state of layer 20 (mid-late), standardize it, and score it with an L2-regularized logistic regression ($C = 0.5$, balanced class weights). The decision threshold is calibrated to an explicit false-positive-rate (FPR) budget (no more than 1%) using *out-of-fold* probabilities, so it matches production scoring rather than an optimistic in-sample ROC curve. A v2 extension concatenates pooled vectors from multiple layers (12, 16, 20, 24). The saved artifact carries a model/layer fingerprint and *refuses to load* on any model or layer mismatch, so a stale probe cannot silently disable the layer.

C. LAYER 3 — OUTPUT AUDITOR

L3 audits the generated response. A cryptographically random *canary* token is embedded in the system prompt and its appearance in the output is treated as system-prompt exfiltration; structural regex heuristics flag compromise phrasing; and a hardened *self-judge* re-examines flagged outputs. The self-judge reuses the already-loaded target LLM (flat VRAM, no second model to OOM), frames the audited text explicitly as data rather than instructions, and parses only the first verdict token so a compromised response cannot suffix-flip the decision.

D. DEPLOYMENT CONSTRAINTS

The system targets two 16 GiB T4 GPUs (Turing, compute capability 7.5): fp16 is forced (bf16 would emulate in software), the 7B model is sharded with `device_map=auto` and per-GPU

memory caps, the surface classifier is pinned to GPU 1, and expandable CUDA segments avoid fragmentation OOMs. The design reuses the single loaded 7B for generation *and* judging, keeping the whole pipeline within the two-T4 budget. The marginal cost of L2 is one extra 7B forward pass per non-blocked request (~50–150 ms at up to 1024 tokens).

IV. EXPERIMENTAL SETUP

A. DATA

Table II lists the corpora. The probe is trained on the **train** split of Deepset’s prompt-injections dataset (546 rows), augmented with 527 jailbreak positives from jailbreak-classification and 339 hard negatives from NotInject [6]—benign text deliberately stuffed with attack trigger words, the direct data-side antidote to over-defense. All evaluation numbers in this paper are computed on the **held-out test split** of prompt-injections (116 rows: 60 injections, 56 benign), which is never seen during training; ChatGPT-Jailbreak-Prompts is reserved for cross-family recall checks.

TABLE II CORPORA USED FOR TRAINING AND HELD-OUT EVALUATION

Corpus	Split	Rows	Semantics
deepset/prompt-injections	train	546	binary labels (203 positive)
deepset/prompt-injections	test	116	held-out (60 inj. / 56 benign)
jailbreak-classification	train	527 pos.	labeled via “type” column
NotInject [6]	3 splits	339	benign + trigger words (hard negatives)
ChatGPT-Jailbreak-Prompts	train	79	all positive (cross-family)

B. TRAINING AND CALIBRATION METHODOLOGY

Three methodological guards matter as much as the fit itself, and each was introduced in response to a measured failure during development. (1) **Evasion augmentation**: each positive is expanded with delimiter smuggling, a base64 payload, and up to seven surface-evasion variants (zero-width characters, homoglyphs, leetspeak, ROT13, HTML comments, fake role tags, whitespace splits); each negative is expanded with a “quoted attack” hard negative so the probe learns the instruction wrapper rather than keywords. (2) **Group-aware cross-validation**: augmented variants of one source row are near-duplicates. With plain stratified folds they straddled fold boundaries and inflated AUC toward 1.0 (measured 0.78 vs. 0.66 AUC on synthetic data with the same structure). Variants now carry a group id and StratifiedGroupKFold keeps each family inside one fold. (3) **Out-of-fold threshold calibration**: reading the FPR budget off an in-sample ROC curve selected a threshold stricter than the budget actually allows (0.9981 vs. 0.9714 in one run), silently costing recall. The threshold is now the largest value whose *out-of-fold* FPR stays within the budget.

C. EVALUATION PROTOCOL

We report true-positive rate (TPR; recall on the 60 held-out injections), FPR (fraction of the 56 benign rows blocked),

accuracy, and ROC-AUC. Two gate policies are compared in one pass over the held-out set: **Policy A (legacy gate)** blocks iff L1 hard-blocks *or* (L1 score is at least $\tau_{\text{gate}} = 0.30$ *and* L2 is at least the threshold); **Policy B (always-on)** blocks iff L1 hard-blocks *or* L2 is at least the threshold. Because the test split is small, we emphasize that FPR has a granularity of 1/56 (~1.8%) per row and we report counts alongside rates; bootstrap confidence intervals are available in the released benchmark harness.

V. RESULTS

A. LAYERS MEASURED IN ISOLATION

Measured independently, the two input layers look like a healthy complement. L1 (the surface classifier) alone blocks 21/60 injections (TPR 0.350) with zero false positives—it catches explicit instructions but is blind to the obfuscated and implicit ones. The L2 hidden-state probe, scored on every row without gating, attains ROC-AUC 0.997, TPR 0.783 at its calibrated threshold, and FPR 0.000. The layers are complementary: of the 39 injections L1 misses, the probe sees most of them clearly. On component metrics alone, adding L2 to L1 should be a large win.

B. THE ROUTING PATHOLOGY

It is not. Table III reports end-to-end behavior. Under Policy A, only 1 of 60 injections ever reaches L2: 21 are hard-blocked by L1 (which is fine), but 38 score below 0.30 at L1 and are silently dropped before the probe can see them. The probe—0.997 AUC in isolation—contributes a single detection, and the ensemble recall (0.367) is indistinguishable from L1 alone (0.350). Lowering or raising L1’s block threshold does not help: the 35 missed injections that score below 0.01 at L1 are simply not in the gate’s forward set, whatever the threshold.

TABLE III END-TO-END GATE-POLICY COMPARISON ON THE HELD-OUT TEST SPLIT (60 INJECTIONS, 56 BENIGN)

Configuration	L2 shown	TPR	FPR	Acc.
L1 only	—	0.350 (21/60)	0.000	0.664
Legacy gate ($\tau_{\text{gate}}=0.30$)	1/60	0.367 (22/60)	0.000	0.672
Always-on, L2 thr 0.9868	39/60	0.833 (50/60)	0.000 (0/56)	0.914
Always-on, L2 thr 0.90	39/60	0.900 (54/60)	0.018 (1/56)	0.940

Making L2 always-on (Policy B) removes the gate and lets the probe see every input L1 does not hard-block. At the artifact’s FPR-budgeted threshold (0.9868), end-to-end recall jumps to 0.833 at zero measured FPR; at the deployed operating point (0.90) it reaches 0.900 at 1.8% FPR. L2 now contributes 29 detections beyond L1’s 21 (50 union). The same signature reproduces on an independent CPU stand-in (SmolLM2-135M with a probe trained on its layer 20): recall 0.44 under the legacy gate versus 0.67 always-on, with L2 adding 0 versus 2 detections beyond L1.

C. OPERATING POINT

Table IV sweeps the probe threshold under always-on routing. FPR is flat (1/56) from 0.50 to 0.95 because only one benign

row ever scores above 0.5 on this split; on broader traffic lower thresholds will cost more false positives. We deploy 0.90 as the balanced point; practitioners whose objective is maximum recall may prefer 0.75–0.50.

TABLE IV THRESHOLD SWEEP UNDER ALWAYS-ON ROUTING (HELD-OUT SPLIT)

L2 threshold	TPR	FPR	Acc.
0.9868 (artifact)	0.833 (50/60)	0.000 (0/56)	0.914
0.95	0.900 (54/60)	0.018 (1/56)	0.940
0.90 (deployed)	0.900 (54/60)	0.018 (1/56)	0.940
0.75	0.933 (56/60)	0.018 (1/56)	0.957
0.50	0.950 (57/60)	0.018 (1/56)	0.966

D. COST OF THE FIX

Always-on routing costs one additional 7B forward pass per non-blocked request (~50–150 ms at up to 1024 tokens on two T4s). VRAM stays flat: L3 reuses the loaded target model for judging, so no second model is introduced. For applications where that latency is unacceptable, the legacy gate can be retained—but operators should know that on our data it reduces L2’s contribution to approximately nothing.

VI. DISCUSSION

A. WHY COMPONENT METRICS DO NOT COMPOSE

Let R_1 be L1’s recall, and let G be the set of injections the gate forwards to L2. If L2 achieves recall R_2 on G , end-to-end recall is

$$R = R_1 + (1 - R_1) \cdot g \cdot R_2$$

where $g = P(\text{input is in } G \mid \text{injection, not blocked by L1})$ is the gate’s coverage of L1’s misses. When the gate forwards only inputs the surface classifier already finds suspicious, g is determined by L1’s blind spots, not by L2’s quality. In our legacy configuration $g = 1/39$ (~0.026), so $R \sim R_1$ regardless of R_2 —a probe with 0.997 AUC contributes a single catch because the gate inherits exactly the inputs L1 cannot see. This is the cascade-recall bound in disguise [13], [14]: no downstream stage can detect what a prior stage refuses to pass on.

B. THREATS TO VALIDITY

We state limitations explicitly because they bound the claims above. (i) The headline numbers come from a single, small, dated corpus (116 held-out rows); FPR granularity is 1.8% per row, so point estimates are noisy and thresholds 0.50–0.95 are indistinguishable on this split. (ii) The deployed threshold was selected using the test split, a mild form of test-set tuning; the released trainer calibrates on a validation slice for a defensible setting. (iii) We do not yet report leave-one-dataset-out generalization or adaptive adversaries (obfuscation, paraphrasing, multi-turn), which the field increasingly requires [10], [11]; the released benchmark harness supports these. (iv) The L1 choice (ProtectAI v2) is a documented over-defense liability [6], and we observe its repetition-triggered false spikes ourselves; the L1 ensemble and dual-key confirmation are mitigations, not eliminations. (v) The L3 self-judge consumes untrusted text and is therefore itself prompt-injectable; it is hardened (data framing, first-token parsing) but scoped

conservatively. We therefore position the contribution as a *measurement of how guardrail stacks fail*, not as a new detector.

C. RECOMMENDATIONS

Our findings translate into concrete practice. (1) *Measure ensembles end-to-end*: report per-layer metrics only as diagnostics; the number that decides a deployment is the ensemble’s recall on the full routing path. (2) *The cheap first stage must have near-total recall, or be bypassable*. If a surface guard’s recall is 0.35, any layer gated behind it inherits a 0.65 ceiling it can never exceed. (3) *Prefer always-on or learned deferral* for expensive stages: the always-on policy here bought +0.53 recall for ~100 ms. (4) *Calibrate thresholds on a validation slice, never the test split, and report confidence intervals* on small sets. (5) *Use over-defense sets* (NotInject) as the FPR benchmark that decides real deployments, since trigger-word bias is the dominant false-positive source.

VII. CONCLUSION

We built and released a three-layer, fail-closed prompt-injection guard for Qwen2.5-7B-Instruct on two T4 GPUs and evaluated it end-to-end. The central, quantified finding is that a hidden-state probe with 0.997 ROC-AUC contributed a single detection when placed behind a conventional escalation gate, because the gate inherited the surface classifier’s blind spots and starved the probe of 38 of 60 held-out injections. Making the second layer always-on raised end-to-end recall from 0.37 to 0.90 at 1.8% FPR. The lesson generalizes: in layered LLM guardrails, the *routing policy* between layers is a first-class security variable, and component metrics do not predict ensemble security. We hope the released pipeline and its failure analysis help practitioners avoid the silent, component-metrics-driven false confidence we measured.

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